



A Beta-Neutral Crypto ML Strategy: From Signal Inversion to Robust Alpha

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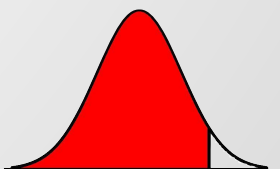
Machine Learning and FinTech

<https://github.com/HWTeng-Teaching/202509-ML-FinTech>

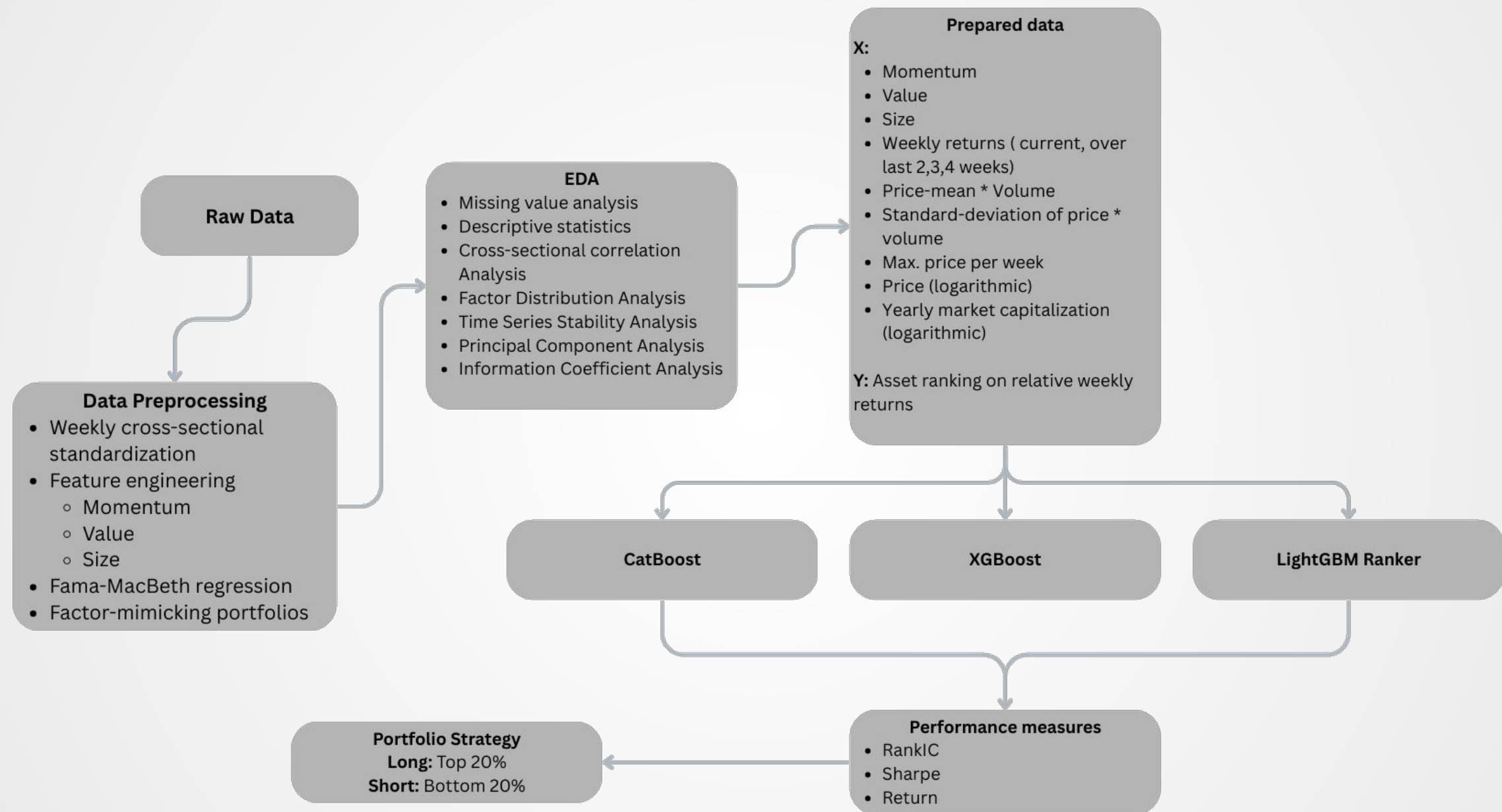


Outline

- ▣ Motivation
- ▣ Data Preprocessing
- ▣ Factor Engineering & EDA
- ▣ Cross-Sectional ML Prediction
- ▣ Portfolio Construction
- ▣ Results & Interpretation



Outline



Motivation

▣ Market Problem

- ▶ Crypto markets exhibit extreme volatility; single assets like BTC frequently suffer **-50% to -70%** drawdowns.
- ▶ Traditional long-only BTC or index exposure delivers **high returns** in bull markets but **catastrophic losses** in bear markets.
- ▶ Institutional and professional investors need **stable, risk-controlled** strategies with superior risk-adjusted returns to BTC.



Motivation

▣ Research Objectives

- ▶ Build a **beta-neutral, long–short, cross-sectional ML strategy** that captures relative performance (alpha) while **minimizing market-direction exposure (beta)**.
- ▶ Control risk: reduce annual volatility and max drawdown to institutionally acceptable levels.
- ▶ Improve Sharpe: target a long-run Sharpe ratio above **2**.
- ▶ Explain alpha sources: use feature importance and Rank IC to identify which factors drive excess returns (mean reversion, quality, liquidity, etc.).



Data Preprocessing

▣ Data Sources & Universe

- ▶ Weekly OHLCV prices and base factors (Binance Spot Market)
- ▶ Market factor: crypto market index CMKT (Binance Spot Market)
- ▶ Annual universe: selecting the **top 30** large-cap / liquid assets per year.
- ▶ 4-week Treasury Bill rate (DTB4WK)
- ▶ Build dataset with **weekly frequency**
- ▶ Keep only the intersection of annual **top 30** and complete **weekly** data to ensure tradability and data quality.



Data Preprocessing

▣ Exploratory Data Analysis

- ▶ Missing value analysis
- ▶ Descriptive statistics
- ▶ Cross-sectional correlation Analysis
- ▶ Factor Distribution Analysis
- ▶ Time Series Stability Analysis
- ▶ Principal Component Analysis
- ▶ Information Coefficient Analysis



Data Preprocessing

▣ Returns & Volatility Definitions

► Weekly excess return:

$$excess_{i,t} = ret_{i,t} - rf_t$$

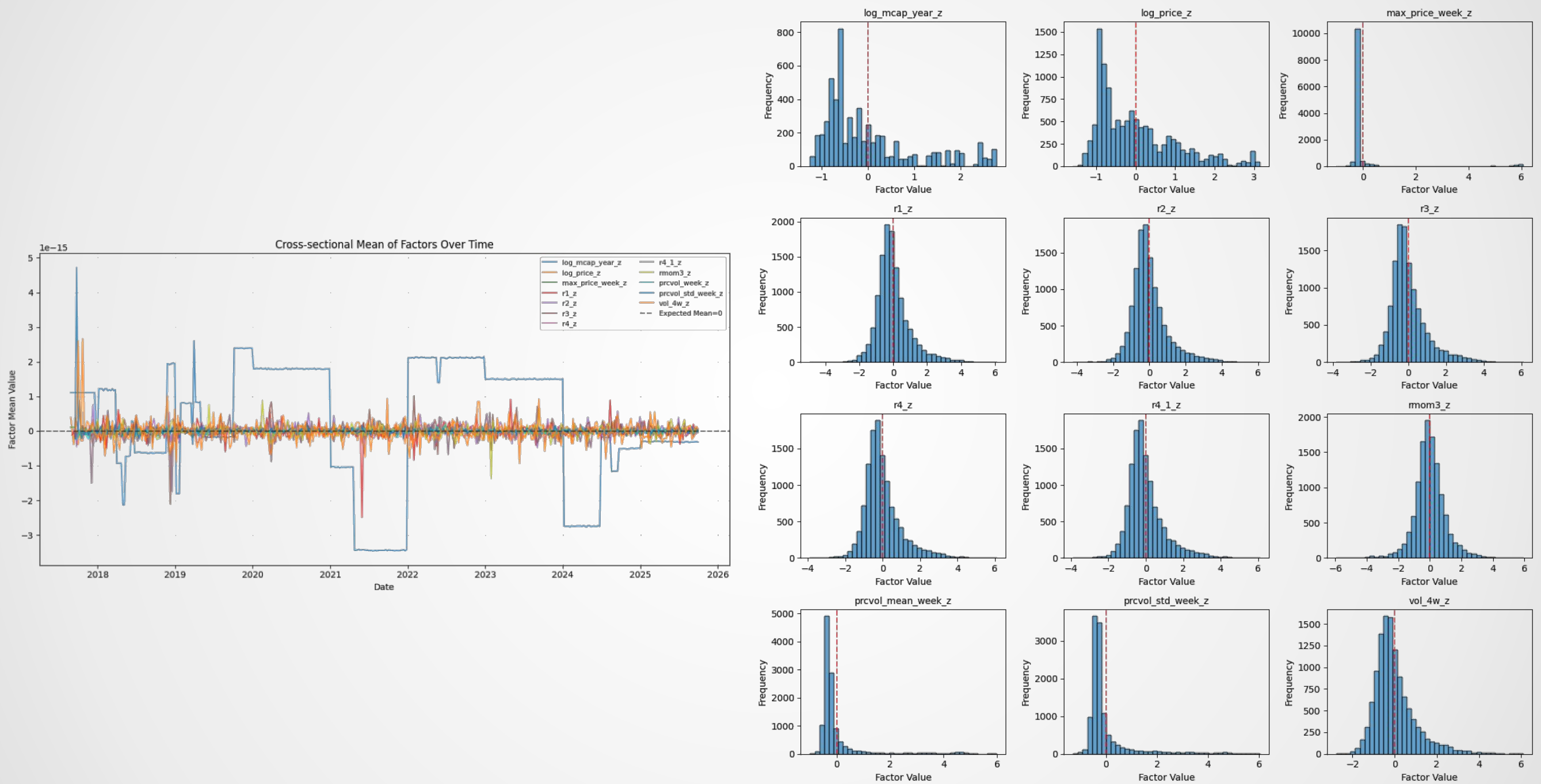
► Next-week excess return (base prediction target):

$$y_{fwd1,i,t} = excess_{i,t} + 1$$

► Risk factors: 4-week and 12-week historical volatility (rolling std of weekly returns).



Data Analysis



Feature Engineering & EDA

▣ Technical & Risk Features

- ▶ We extract from weekly OHLCV :
- ▶ RSI (14): overbought / oversold conditions.
- ▶ ATR (14) / Price: relative volatility.
- ▶ Bollinger Band Width (20): range-based volatility.
- ▶ MACD Histogram: trend momentum shifts.
- ▶ Price / MA(20, 50): deviation from moving averages.
- ▶ Rolling Volatility (4w, 12w): short- and medium-term risk.
- ▶ All features are **cross-sectionally z-scored per week**, focusing on **relative levels** within each week and enabling cross-sectional ranking.



Feature Engineering & EDA

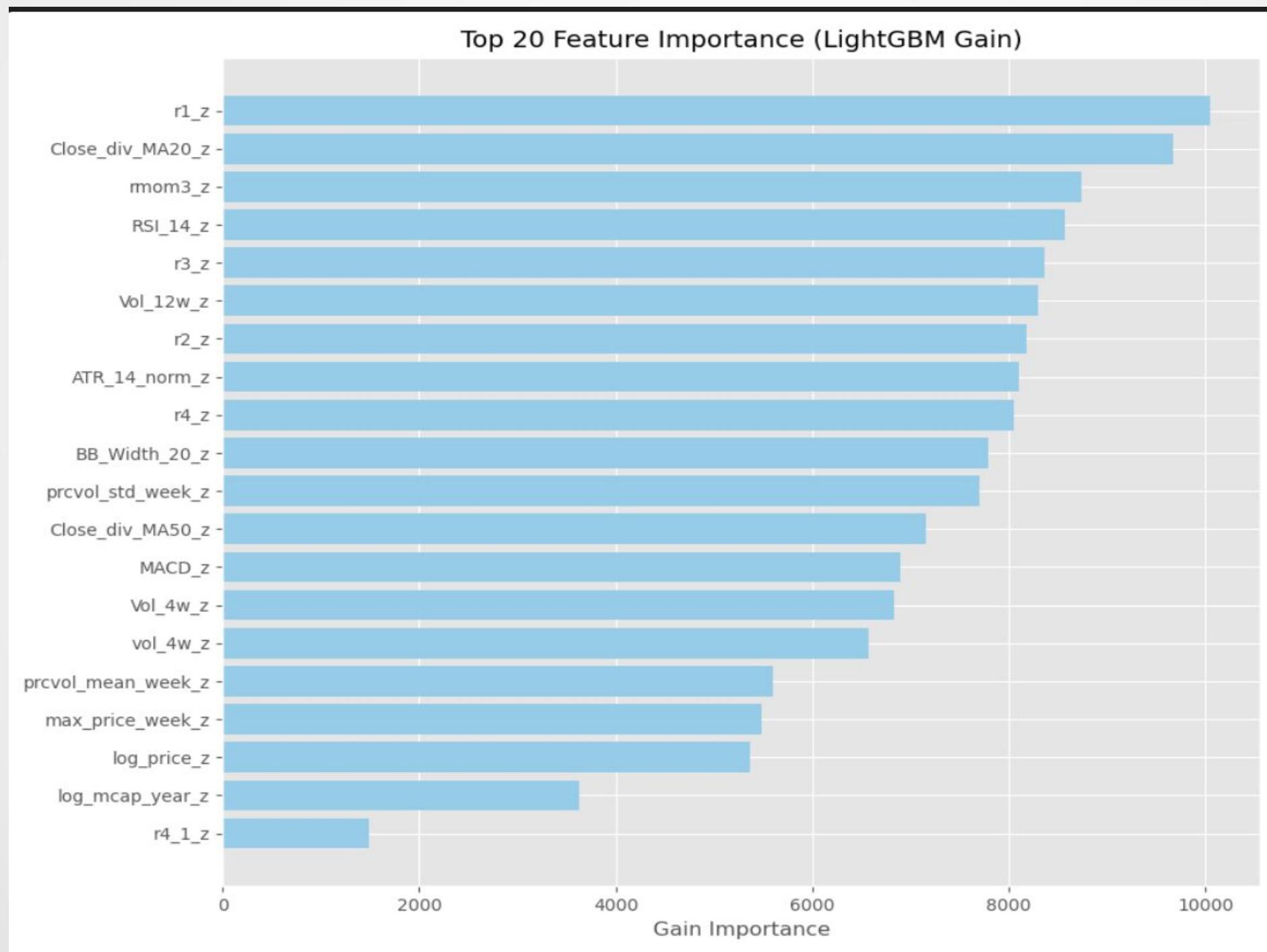
▣ Feature Importance

- ▶ **Top-20** LightGBM Gain features are dominated by:
- ▶ Short-/medium-term returns: r1_z, r2_z, r3_z, r4_z, mom3_z.
- ▶ Technical indicators: Close_div_MA20_z, RSI_14_z, Close_div_MA50_z, MACD_z.
- ▶ Volatility & ranges: Vol_12w_z, Vol_4w_z, ATR_14_norm_z, BB_Width_20_z.
- ▶ Liquidity/volume: prcvol_mean_week_z, prcvol_std_week_z.
- ▶ Conclusion: the model heavily relies on “**short-term momentum + technical + volatility/volume**” signals, while size/price play a supporting backbone role.



Feature Engineering & EDA

▣ Feature Importance



Feature Engineering & EDA

▣ Rank IC Evidence

► Factors with **positive** Mean Rank IC:

► log_mcap_year_z, log_price_z, max_price_week_z, prcvol_mean_week_z, prcvol_std_week_z.

► ⇒ **Large-cap, higher-priced and more liquid/active assets** tend to have better **risk-adjusted returns** (quality / liquidity tilt).

► Factors with **negative** Mean Rank IC:

► Vol_4w_z, Vol_12w_z, ATR_14_norm_z, BB_Width_20_z.

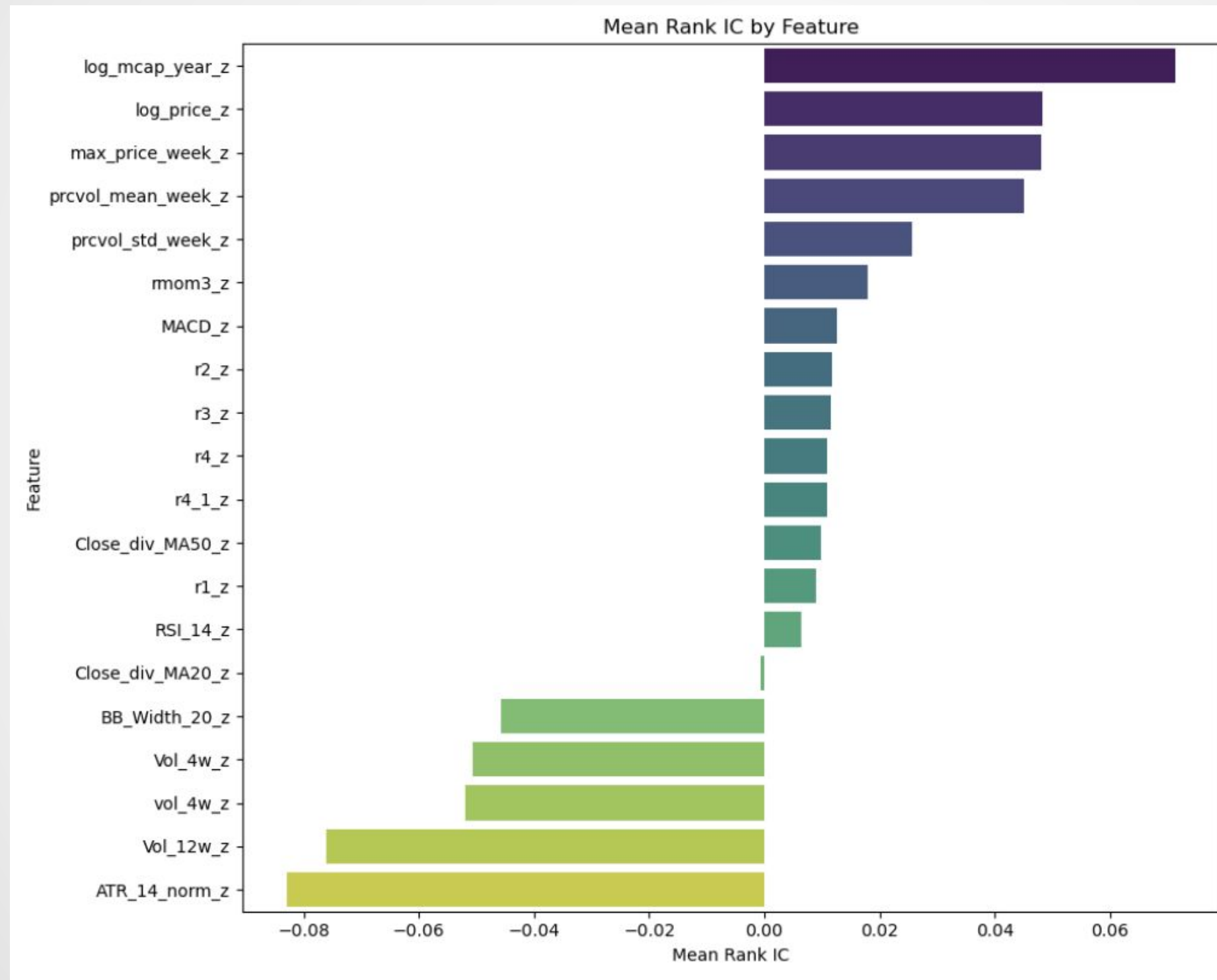
► ⇒ **High-volatility, high-ATR assets** typically **underperform**, making them good **short** or **de-weighting** candidates.

► Cumulative Rank IC shows that **size/liquidity factors** accumulate IC over time, indicating persistent structural alpha rather than one-off episodes



Feature Engineering & EDA

Rank IC Evidence



Cross-Sectional ML Prediction

▣ Volatility-Scaled Target

► Prediction target:

$$\text{Target}_{i,t} = \frac{y_fwd1, i, t}{\text{Vol_}4w, i, t}$$

► Prevent the model from simply chasing high-volatility, high-return small tokens.

► Force the model to learn **risk-adjusted excess returns**, improving robustness.



Cross-Sectional ML Prediction

▣ DART Boosting / Model Architecture & DART

- ▶ Main model: LightGBM LambdaRank (cross-sectional ranking)
- ▶ DART = Dropouts meet Additive Regression Trees.
- ▶ Randomly drops part of the existing trees before fitting a new one
- ▶ Benefits: reduces overfitting and improves generalization, especially on noisy financial data.
- ▶ Auxiliary model: Ridge Regression
- ▶ Provides a linear, smooth baseline and mitigates extreme nonlinear biases.
- ▶ Ensemble: 0.6 LGB + 0.4 Ridge



Cross-Sectional ML Prediction

▣ Rolling Walk-Forward Setup

- ▶ Use at least **52 weeks** of history as the initial **training set**; the most recent **8 weeks** are used for **validation**.
- ▶ Refit the model every 4 weeks, processing a total of 373 weeks of data.
- ▶ Each week is treated as a ranking group; the model learns which assets should rank higher or lower within the same week.



Portfolio Construction

▣ Bucketing & Signal Inversion

- ▶ Each week, ensemble scores are bucketed into 5 quantiles ($Q=5$).
- ▶ raw scores are **negatively** correlated with **future returns**.
- ▶ **Long**: **bottom** 20% (lowest-score quantile).
- ▶ **Short**: **top** 20% (highest-score quantile).
- ▶ Interpretation: the model effectively detects **over-extended** assets; **high scores indicate short-term overheating** and likely **mean reversion**, while low scores correspond to overlooked but higher-quality names.



Portfolio Construction

▣ Risk Constraints

- ▶ Total gross leverage: $\text{GROSS_LEV} = 1.0$ ($|\text{long}| + |\text{short}| = 100\%$).
- ▶ Per-asset weight cap: $|w_i| \leq 10\%$.
- ▶ Weekly per-asset turnover cap: $|\Delta w_i| \leq 30\%$.
- ▶ One-way transaction cost: 10 bps.
- ▶ Goal: control concentration and trading frictions, making the strategy implementable in practice.



Portfolio Construction

▣ Beta Neutrality & Vol Targeting

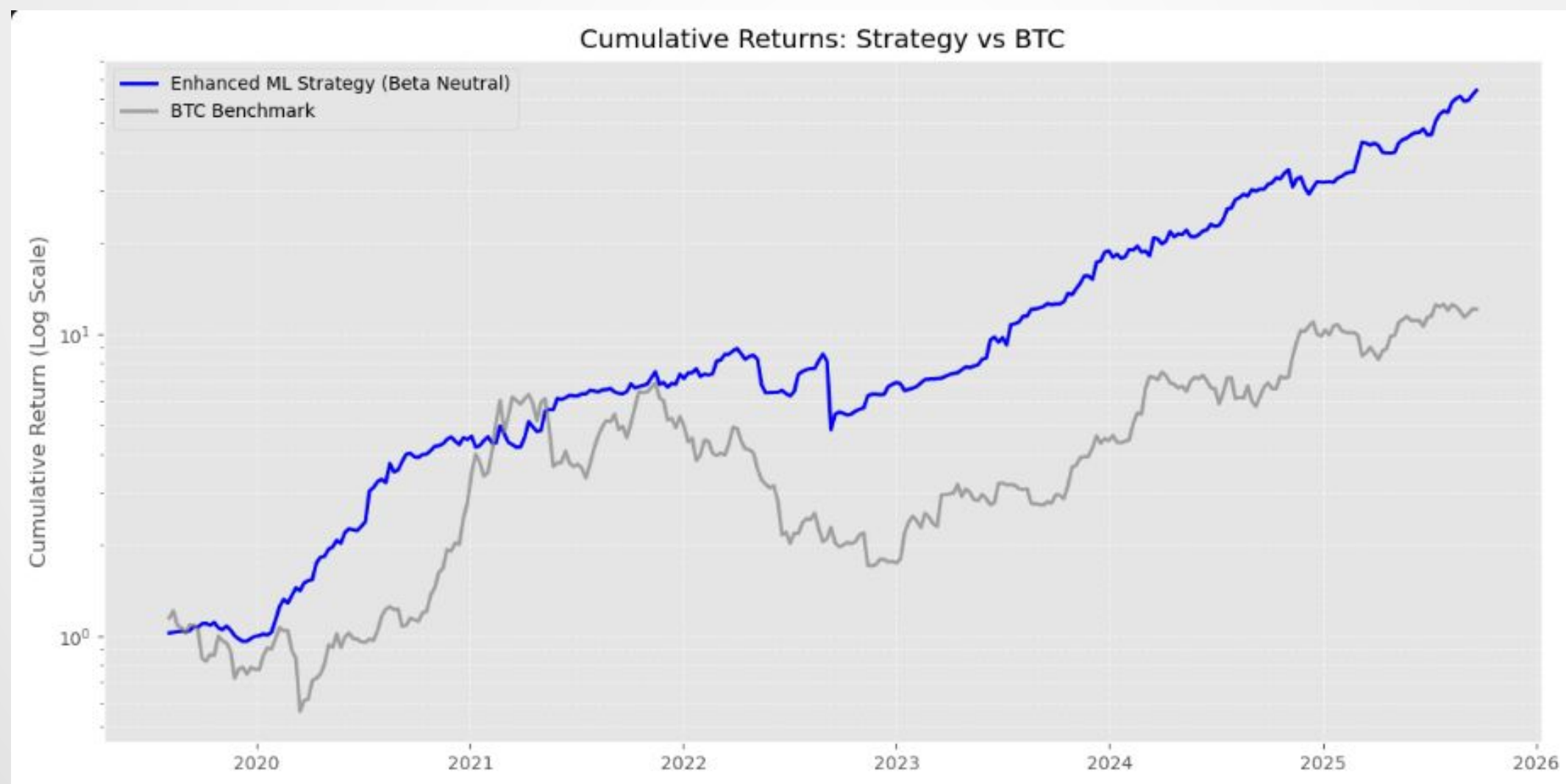
- ▶ Estimate each asset's beta via a 52-week rolling regression on CMKT.
- ▶ Adjust portfolio weights by
- ▶ to achieve near **beta neutrality**.
- ▶ Using 26-week historical volatility, scale exposure to keep annualized volatility around **30%**:
- ▶ **De-leverage** in **high-vol** regimes, **scale up** in **low-vol** regimes.
- ▶ Result: a stable and more investable risk profile.



Results & Interpretation

▣ Cumulative returns (log scale)

- ▶ The beta-neutral strategy (blue) shows a **smooth**, steadily rising curve, clearly outperforming BTC (gray) over the full sample.
- ▶ Log scale allows fair comparison across different compounding phases.



Results & Interpretation

▣ ML Strategy VS BTC Benchmark

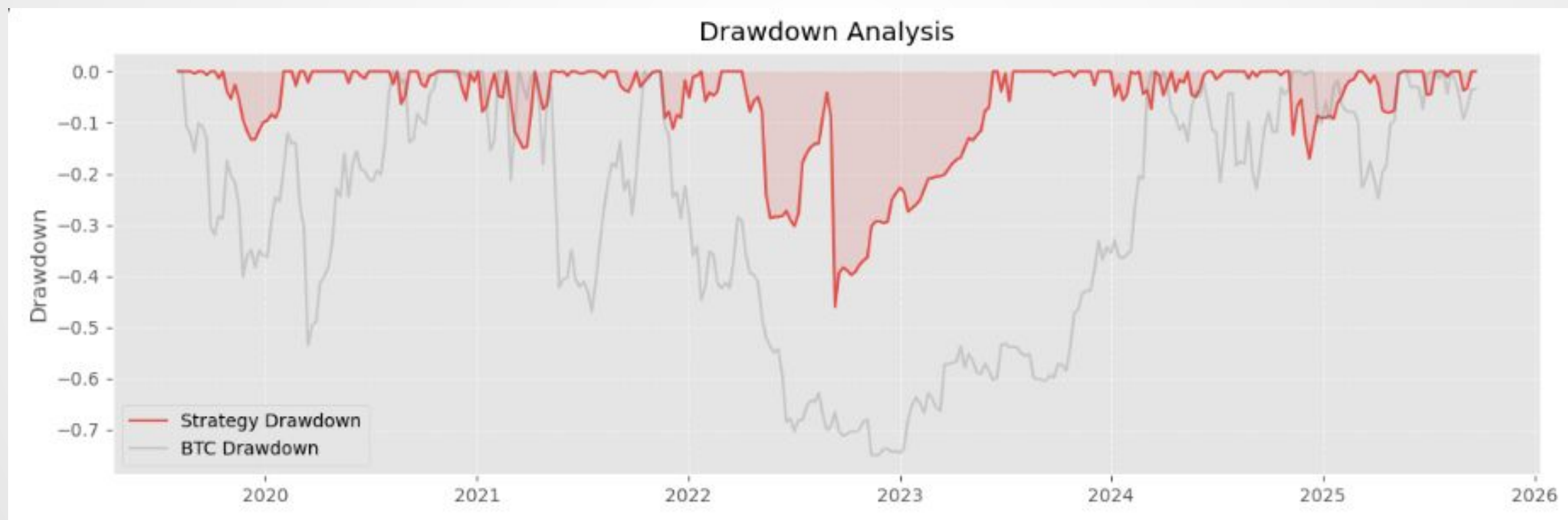
	Benchmark	Strategy
Annual Return	3831%	75%
Annual Volatility	9577%	37%
Sharpe Ratio	0.4	2.03
Max Drawdown	-70.1%	-46.1%
Avg Turnover	0.72	0.72



Results & Interpretation

▣ Drawdown

- ▶ Strategy max drawdown is around **-20% to -25%**, with most periods within **-10%**.
- ▶ BTC experiences multiple drawdowns beyond **-50%**, approaching **-70%**.



Results & Interpretation

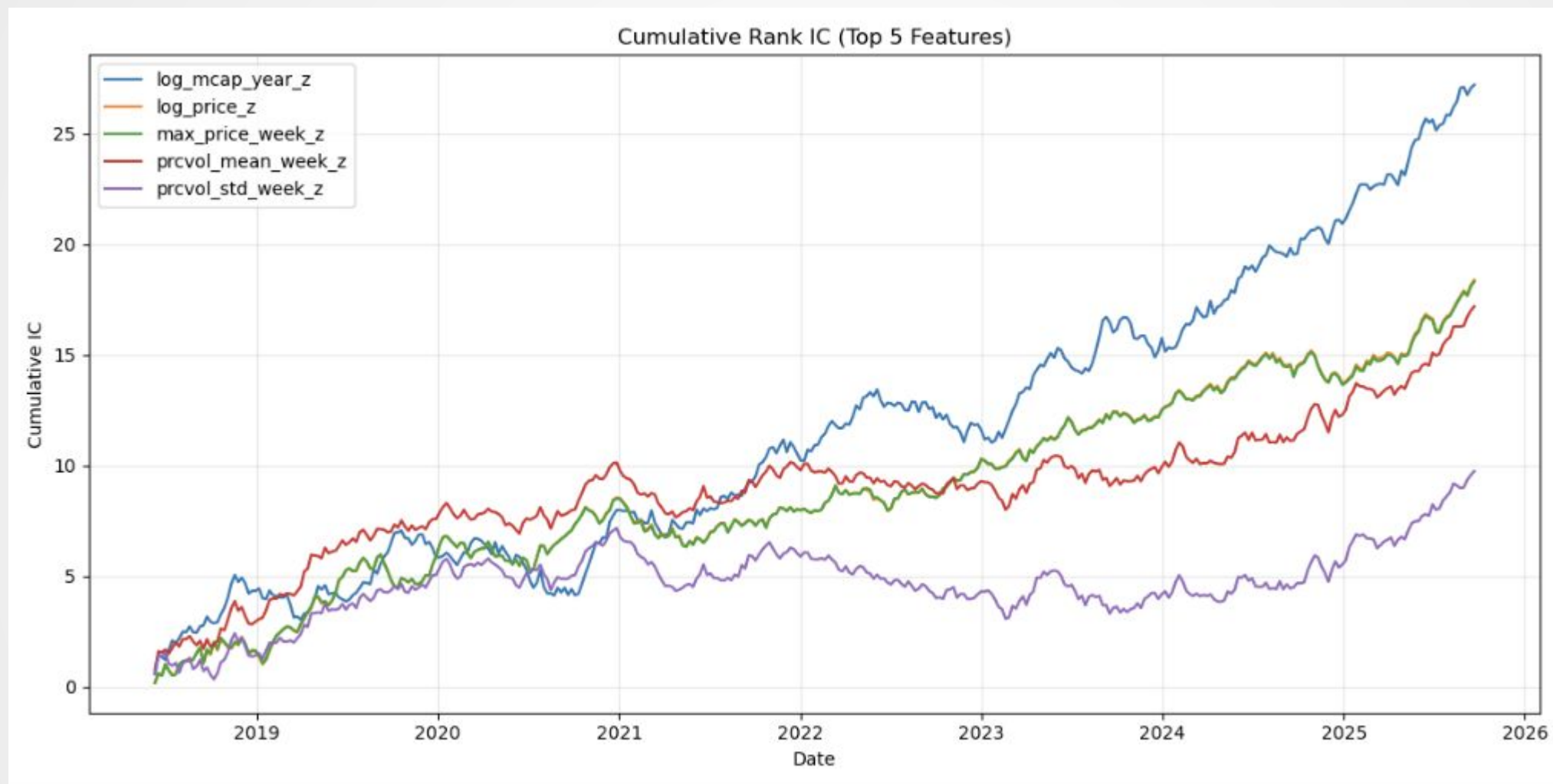
▣ Feature Importance & Signal Inversion

- ▶ **LightGBM** primarily relies on momentum + technical + volatility features:
- ▶ **High-scoring** assets are typically “recent high flyers with elevated volatility and overheated technicals.”
- ▶ **Mean Rank IC** shows volatility and ATR-type features carry **negative** signals:
- ▶ The more volatile the asset, the worse it tends to perform next week.
- ▶ ⇒ **Over-extended assets likely to underperform,**
- ▶ Inverting the signal (long low scores, short high scores) with a **mean-reversion** strategy.



Results & Interpretation

▣ Cumulative Rank IC



Results & Interpretation

▣ Cumulative Rank IC

- ▶ Large-cap, liquid, high-attention assets consistently outperform on a risk-adjusted basis, creating a **structural quality/liquidity alpha**.
- ▶ The strategy effectively combines two core components:
- ▶ A **quality/liquidity tilt** toward **large, liquid names**.
- ▶ A **mean-reversion** short/underweight leg against **short-term over-extended, high-volatility names**.



Results & Interpretation

▣ Conclusion

- ▶ Starting from the motivation: in a noisy, highly volatile crypto market, we aim for a risk-controlled, beta-neutral alpha strategy.
- ▶ Through data filtering and feature engineering, we focus on the annual top-30 large-cap universe and build momentum, technical, volatility, liquidity, and size factors.
- ▶ Using a volatility-scaled target with DART-LightGBM and Ridge ensemble, we learn which assets will outperform on a risk-adjusted basis in the next week.
- ▶ Empirically, model scores detect “over-extended vs stable high-quality” assets, which naturally leads to signal inversion plus long–short, beta neutrality, and volatility targeting.
- ▶ Final outcome:
 - ▶ Cumulative returns consistently outperform BTC;
 - ▶ Annual volatility and max drawdown are greatly reduced;
 - ▶ Sharpe ≈ 2 with a clear, interpretable strategy (mean reversion + quality/liquidity).





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